CREATING DATABASE & SETTING UP THE REQUIRED TABLES

STEP 1: CREATING & USING DATABASE

CREATE DATABASE healthcare_analytics;
USE healthcare_analytics;

```
MariaDB [(none)]> CREATE DATABASE healthcare_analytics;
Query OK, 1 row affected (0.008 sec)

MariaDB [(none)]> USE healthcare_analytics;
Database changed
```

STEP 2: CORE TABLES

2.1 - users

```
CREATE TABLE users (
   user_id    INT AUTO_INCREMENT PRIMARY KEY,
   username   VARCHAR(100) UNIQUE NOT NULL,
   password   VARCHAR(255) NOT NULL,
   email      VARCHAR(255) UNIQUE NOT NULL,
   role       ENUM('admin', 'doctor', 'patient', 'analyst') NOT NULL,
   created_on DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE users (
    ->     user_id    INT AUTO_INCREMENT PRIMARY KEY,
    ->     username   VARCHAR(100) UNIQUE NOT NULL,
    ->     password   VARCHAR(255) NOT NULL,
    ->     email      VARCHAR(255) UNIQUE NOT NULL,
    ->     role       ENUM('admin', 'doctor', 'patient', 'analyst') NOT NULL,
    ->     created_on DATETIME DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.066 sec)
```

2.2 - hospitals

```
CREATE TABLE hospitals (
   hospital_id  INT AUTO_INCREMENT PRIMARY KEY,
   name         VARCHAR(255) NOT NULL,
   location     VARCHAR(255),
   contact_info VARCHAR(255)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE hospitals (
    ->     hospital_id  INT AUTO_INCREMENT PRIMARY KEY,
    ->     name         VARCHAR(255) NOT NULL,
    ->     location     VARCHAR(255),
    ->     contact_info VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.019 sec)
```

STEP 3: USER-RELATED TABLE

3.1 - user_profiles

```
CREATE TABLE user_profiles (
   profile_id     INT AUTO_INCREMENT PRIMARY KEY,
   user_id        INT NOT NULL,
   first_name     VARCHAR(100),
   last_name      VARCHAR(100),
   date_of_birth  DATE,
   contact_number VARCHAR(20),
   address        TEXT,
   FOREIGN KEY (user_id) REFERENCES users(user_id)
      ON DELETE CASCADE
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE user_profiles (
    ->     profile_id     INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id        INT NOT NULL,
    ->     first_name     VARCHAR(100),
    ->     last_name      VARCHAR(100),
    ->     date_of_birth  DATE,
    ->     contact_number VARCHAR(20),
    ->     address        TEXT,
    ->     FOREIGN KEY (user_id) REFERENCES users(user_id)
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.031 sec)
```

3.2 - patients

```
CREATE TABLE patients (
   patient_id        INT AUTO_INCREMENT PRIMARY KEY,
   user_id           INT NOT NULL,
   gender            VARCHAR(10),
   blood_group       VARCHAR(10),
   emergency_contact VARCHAR(20),
   FOREIGN KEY (user_id) REFERENCES users(user_id)
      ON DELETE CASCADE
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE patients (
    ->     patient_id        INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id           INT NOT NULL,
    ->     gender            VARCHAR(10),
    ->     blood_group       VARCHAR(10),
    ->     emergency_contact VARCHAR(20),
    ->     FOREIGN KEY (user_id) REFERENCES users(user_id)
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.039 sec)
```

3.3 - doctors

```
CREATE TABLE doctors (
   doctor_id      INT AUTO_INCREMENT PRIMARY KEY,
   user_id        INT NOT NULL,
   hospital_id    INT,
   specialization VARCHAR(100),
   license_number VARCHAR(50) UNIQUE,
   FOREIGN KEY (user_id) REFERENCES users(user_id)
      ON DELETE CASCADE,
   FOREIGN KEY (hospital_id) REFERENCES hospitals(hospital_id)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE doctors (
    ->     doctor_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id        INT NOT NULL,
    ->     hospital_id    INT,
    ->     specialization VARCHAR(100),
    ->     license_number VARCHAR(50) UNIQUE,
    ->     FOREIGN KEY (user_id) REFERENCES users(user_id)
    ->         ON DELETE CASCADE,
    ->     FOREIGN KEY (hospital_id) REFERENCES hospitals(hospital_id)
    -> );
Query OK, 0 rows affected (0.038 sec)
```

STEP 4: APPOINTMENT MODULE

4.1 - appointment

```
CREATE TABLE appointments (
    appointment_id   INT AUTO_INCREMENT PRIMARY KEY,
    patient_id       INT NOT NULL,
    doctor_id        INT NOT NULL,
    hospital_id      INT,
    appointment_date DATE,
    status           ENUM('scheduled', 'completed', 'cancelled'),
    reason           TEXT,
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id),
    FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id),
    FOREIGN KEY (hospital_id) REFERENCES hospitals(hospital_id)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE appointments (
    ->     appointment_id   INT AUTO_INCREMENT PRIMARY KEY,
    ->     patient_id       INT NOT NULL,
    ->     doctor_id        INT NOT NULL,
    ->     hospital_id      INT,
    ->     appointment_date DATE,
    ->     status           ENUM('scheduled', 'completed', 'cancelled'),
    ->     reason           TEXT,
    ->     FOREIGN KEY (patient_id) REFERENCES patients(patient_id),
    ->     FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id),
    ->     FOREIGN KEY (hospital_id) REFERENCES hospitals(hospital_id)
    -> );
Query OK, 0 rows affected (0.042 sec)
```

STEP 5: MEDICAL DATA TABLES

5.1 - patient_vitals

```
CREATE TABLE patient_vitals (
    vital_id         INT AUTO_INCREMENT PRIMARY KEY,
    patient_id       INT NOT NULL,
    recorded_at      DATETIME DEFAULT CURRENT_TIMESTAMP,
    heart_rate       INT,
    bp_systolic      INT,
    bp_diastolic     INT,
    temperature      FLOAT,
    respiratory_rate INT,
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE patient_vitals (
    ->     vital_id         INT AUTO_INCREMENT PRIMARY KEY,
    ->     patient_id       INT NOT NULL,
    ->     recorded_at      DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     heart_rate       INT,
    ->     bp_systolic      INT,
    ->     bp_diastolic     INT,
    ->     temperature      FLOAT,
    ->     respiratory_rate INT,
    ->     FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
    -> );
Query OK, 0 rows affected (0.040 sec)
```

5.2 - medical_records

```
CREATE TABLE medical_records (
    record_id      INT AUTO_INCREMENT PRIMARY KEY,
    patient_id     INT NOT NULL,
    doctor_id      INT NOT NULL,
    hospital_id    INT,
    visit_date     DATE,
    diagnosis      TEXT,
    treatment_plan TEXT,
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id),
    FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id),
    FOREIGN KEY (hospital_id) REFERENCES hospitals(hospital_id)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE medical_records (
    ->     record_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     patient_id     INT NOT NULL,
    ->     doctor_id      INT NOT NULL,
    ->     hospital_id    INT,
    ->     visit_date     DATE,
    ->     diagnosis      TEXT,
    ->     treatment_plan TEXT,
    ->     FOREIGN KEY (patient_id) REFERENCES patients(patient_id),
    ->     FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id),
    ->     FOREIGN KEY (hospital_id) REFERENCES hospitals(hospital_id)
    -> );
Query OK, 0 rows affected (0.047 sec)
```

5.3 - prescriptions

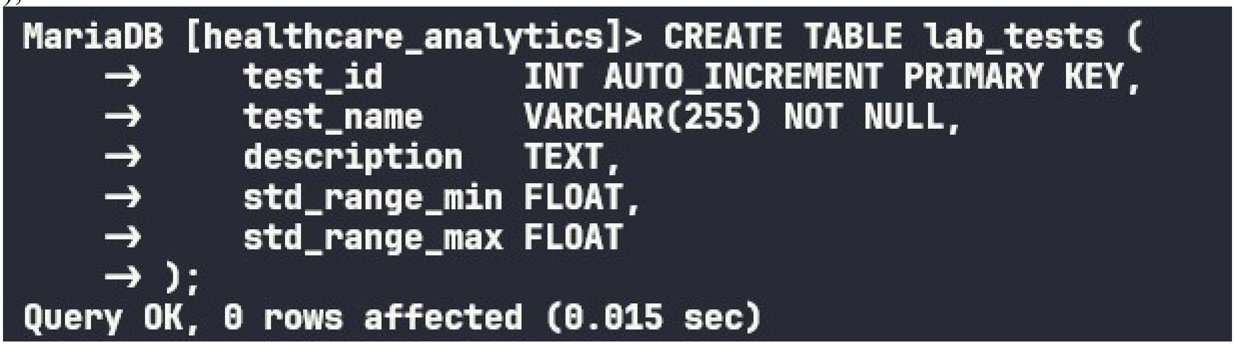
```
CREATE TABLE prescriptions (
    prescription_id INT AUTO_INCREMENT PRIMARY KEY,
    record_id       INT NOT NULL,
    medication_name VARCHAR(255),
    dosage          VARCHAR(100),
    frequency       VARCHAR(100),
    start_date      DATE,
    end_date        DATE,
    FOREIGN KEY (record_id) REFERENCES medical_records(record_id)
        ON DELETE CASCADE
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE prescriptions (
    ->     prescription_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     record_id       INT NOT NULL,
    ->     medication_name VARCHAR(255),
    ->     dosage          VARCHAR(100),
    ->     frequency       VARCHAR(100),
    ->     start_date      DATE,
    ->     end_date        DATE,
    ->     FOREIGN KEY (record_id) REFERENCES medical_records(record_id)
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.038 sec)
```

STEP 6: LAB TESTS

6.1 - lab_tests

```
CREATE TABLE lab_tests (
    test_id       INT AUTO_INCREMENT PRIMARY KEY,
    test_name     VARCHAR(255) NOT NULL,
    description   TEXT,
    std_range_min FLOAT,
    std_range_max FLOAT
);
```



```
MariaDB [healthcare_analytics]> CREATE TABLE lab_tests (
    ->     test_id       INT AUTO_INCREMENT PRIMARY KEY,
    ->     test_name     VARCHAR(255) NOT NULL,
    ->     description   TEXT,
    ->     std_range_min FLOAT,
    ->     std_range_max FLOAT
    -> );
Query OK, 0 rows affected (0.015 sec)
```

STEP 7: BILLING & INSURANCE

7.1 - billings

```
CREATE TABLE billings (
    bill_id        INT AUTO_INCREMENT PRIMARY KEY,
    appointment_id INT NOT NULL,
    patient_id     INT NOT NULL,
    amount         FLOAT,
    payment_status ENUM('paid', 'unpaid', 'pending'),
    generated_at   DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (appointment_id) REFERENCES appointments(appointment_id),
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id));
```

```
MariaDB [healthcare_analytics]> CREATE TABLE billings (
    ->     bill_id        INT AUTO_INCREMENT PRIMARY KEY,
    ->     appointment_id INT NOT NULL,
    ->     patient_id     INT NOT NULL,
    ->     amount         FLOAT,
    ->     payment_status ENUM('paid', 'unpaid', 'pending'),
    ->     generated_at   DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (appointment_id) REFERENCES appointments(appointment_id),
    ->     FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
    -> );
Query OK, 0 rows affected (0.039 sec)
```

7.2 - insurance

```
CREATE TABLE insurance (
    insurance_id     INT AUTO_INCREMENT PRIMARY KEY,
    patient_id       INT NOT NULL,
    provider_name    VARCHAR(255),
    policy_number    VARCHAR(100),
    coverage_details TEXT,
    expiry_date      DATE,
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE insurance (
    ->     insurance_id     INT AUTO_INCREMENT PRIMARY KEY,
    ->     patient_id       INT NOT NULL,
    ->     provider_name    VARCHAR(255),
    ->     policy_number    VARCHAR(100),
    ->     coverage_details TEXT,
    ->     expiry_date      DATE,
    ->     FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
    -> );
Query OK, 0 rows affected (0.032 sec)
```

STEP 8: ANALYTICS & AI TABLES

8.1 - health_risk_scores

```
CREATE TABLE health_risk_scores (
   score_id      INT AUTO_INCREMENT PRIMARY KEY,
   patient_id    INT NOT NULL,
   calculated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
   risk_model    VARCHAR(255),
   score_value   FLOAT,
   risk_level    VARCHAR(50),
   FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE health_risk_scores (
    →     score_id      INT AUTO_INCREMENT PRIMARY KEY,
    →     patient_id    INT NOT NULL,
    →     calculated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    →     risk_model    VARCHAR(255),
    →     score_value   FLOAT,
    →     risk_level    VARCHAR(50),
    →     FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
    → );
Query OK, 0 rows affected (0.037 sec)
```

8.2 - analytics_logs

```
CREATE TABLE analytics_logs (
   log_id          INT AUTO_INCREMENT PRIMARY KEY,
   user_id         INT NOT NULL,
   entity_affected VARCHAR(255),
   timestamp       DATETIME DEFAULT CURRENT_TIMESTAMP,
   details         TEXT,
   FOREIGN KEY (user_id) REFERENCES users(user_id)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE analytics_logs (
    →     log_id          INT AUTO_INCREMENT PRIMARY KEY,
    →     user_id         INT NOT NULL,
    →     entity_affected VARCHAR(255),
    →     timestamp       DATETIME DEFAULT CURRENT_TIMESTAMP,
    →     details         TEXT,
    →     FOREIGN KEY (user_id) REFERENCES users(user_id)
    → );
Query OK, 0 rows affected (0.033 sec)
```

STEP 9: AUDIT LOGS (ADMIN)
9.1 - audit_logs

```
CREATE TABLE audit_logs (
   audit_id    INT AUTO_INCREMENT PRIMARY KEY,
   user_id     INT NOT NULL,
   table_name  VARCHAR(100),
   record_id   INT,
   action      VARCHAR(255),
   old_value   TEXT,
   new_value   TEXT,
   timestamp   DATETIME DEFAULT CURRENT_TIMESTAMP,
   FOREIGN KEY (user_id) REFERENCES users(user_id)
);
```

```
MariaDB [healthcare_analytics]> CREATE TABLE audit_logs (
    ->     audit_id    INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id     INT NOT NULL,
    ->     table_name  VARCHAR(100),
    ->     record_id   INT,
    ->     action      VARCHAR(255),
    ->     old_value   TEXT,
    ->     new_value   TEXT,
    ->     timestamp   DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (user_id) REFERENCES users(user_id)
    -> );
Query OK, 0 rows affected (0.035 sec)
```

STEP 10: FINAL VERIFICATION

SHOW TABLES; -- This will show all the tables in the healthcare_analytics database

```
MariaDB [healthcare_analytics]> SHOW TABLES;
+--------------------------------+
| Tables_in_healthcare_analytics |
+--------------------------------+
| analytics_logs                 |
| appointments                   |
| audit_logs                     |
| billings                       |
| doctors                        |
| health_risk_scores             |
| hospitals                      |
| insurance                      |
| lab_tests                      |
| medical_records                |
| patient_vitals                 |
| patients                       |
| prescriptions                  |
| user_profiles                  |
| users                          |
+--------------------------------+
15 rows in set (0.005 sec)
```